

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and transmission as an assembly without causing personal injury.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

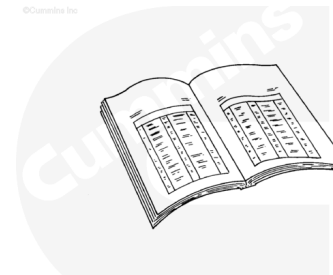
To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Remove the engine and place on an engine stand. Refer to Procedure 000-001 in Section 00.
(/qs3/pubsys2/xml/en/procedures/40/40-000-001.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html)
- Remove the rocker levers. Refer to Procedure 003-008 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html)

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- Remove the push rods. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/40/40-004-014-tr.html>)
- Remove the cooling fan drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html>)
- If required, remove the fan hub pulley. Refer to Procedure 008-039 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html>)
- Remove the vibration damper/crankshaft pulley. If equipped with a viscous damper. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html>) Use the following procedure if equipped with a rubber damper. Refer to Procedure 001-051 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html>) If equipped with a crankshaft pulley only. Refer to Procedure 001-022 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-022-tr.html>)
- Remove the front cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-031-tr.html>)
- Remove the flywheel or flexplate. Refer to Procedure 016-005 (</qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html>) or Refer to Procedure 016-004 in Section 16. (</qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html>)
- Remove the flywheel housing. Refer to Procedure 016-006 in Section 16. (</qs3/pubsys2/xml/en/procedures/40/40-016-006-tr.html>)
- For front gear train engines, remove the rear seal and rear seal carrier. Refer to Procedure 001-024 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-024-tr.html>) Refer to Procedure 001-104 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-104-tr.html>)
- For rear gear train engines, remove the camshaft gear. Refer to Procedure 001-012 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-012-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.

(/qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html)

- If equipped, remove the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-089-tr.html)
- If equipped, remove the balancer. Refer to Procedure 001-004 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-004.html)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html)
- Remove the fuel injection pump. Use the following procedure for rotary fuel injection pumps. Refer to Procedure 005-014 in Section 5. Use the following procedure for inline fuel injection pumps.
(/qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html) Refer to Procedure 005-012 in Section 5.
(/qs3/pubsys2/xml/en/procedures/40/40-005-012-tr.html)
- If equipped, remove or disconnect any driven accessories.
- If equipped, remove the air compressor. Refer to Procedure 012-014 in Section 12.
(/qs3/pubsys2/xml/en/procedures/40/40-012-014-tr.html)
- Remove the fuel transfer pump. Refer to Procedure 005-045 in Section 5.
(/qs3/pubsys2/xml/en/procedures/40/40-005-045-tr.html)

Service Tip: With the engine on the engine stand, rotate the engine so that it is upside down. This will allow the tappets to move off of the camshaft and eliminate the need to use tappet replacement kit, Part Number 3822513, to raise the tappets.

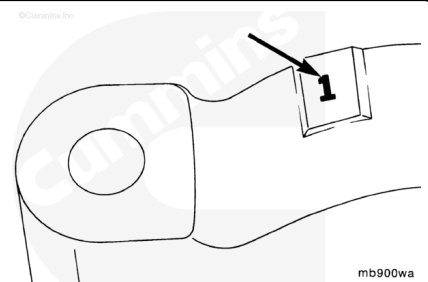
- Remove the camshaft. Refer to Procedure 001-008 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html)
- For rear gear train engines, remove the rear gear housing. Refer to Procedure 001-034 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-034-tr.html)
- For front gear train engines, remove the front gear housing. Refer to Procedure 001-033 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-033-tr.html)

- For front gear train engines, remove the rear seal and rear seal carrier. Refer to Procedure 001-067 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-067.html)
- If equipped, remove the J-jet piston cooling nozzles. Refer to Procedure 001-046 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-046-tr.html)
- Remove the connecting rod caps. Refer to Procedure 001-005 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-005.html)

Remove

Before removing the main bearing caps, make certain that the caps are clearly marked for their location on the lubricating oil cooler side of the main bearing cap and cylinder block.

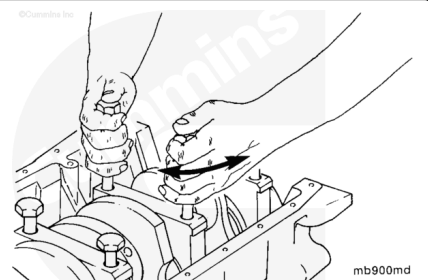
The number one cap is at the front of the engine.



Loosen the main bearing capscrews completely, but do **not** remove.

⚠ CAUTION ⚠

Do not pry on the main bearing caps to free them from the cylinder block. Damage to the main bearing caps and cylinder block can result.



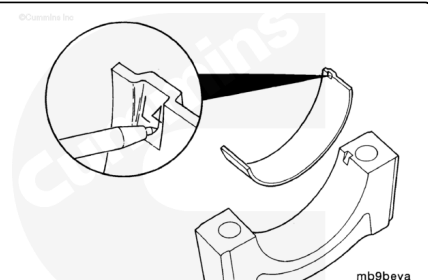
Use two of the main bearing cap bolts to “wiggle” the main bearing cap loose, being careful **not** to damage the bolt threads.

Remove the main bearing cap.

Mark the lower main bearings for position and number as they are removed.

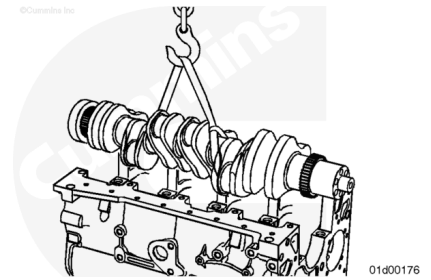
Use an awl to mark the bearing position in the tang area.

Note : Mark the bearing's position for future identification or possible failure analysis.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : Lift the crankshaft straight up to reduce the possibility of damage to the crankshaft and cylinder block.

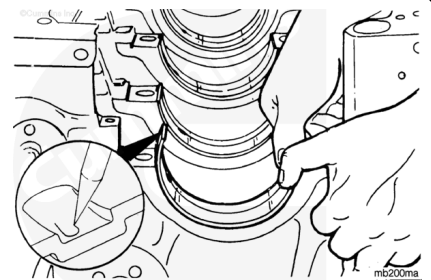
Install nylon lift sling, Part Number 3375957, around Number 3 and Number 4 rod bearing journals.

Attach the sling to a hoist and remove the crankshaft.

Remove the upper main bearings.

Use an awl to mark the bearing's position in the tang area.

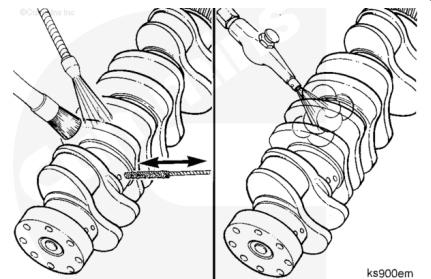
Note : Mark the bearing's position for future identification or possible failure analysis.



Clean

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

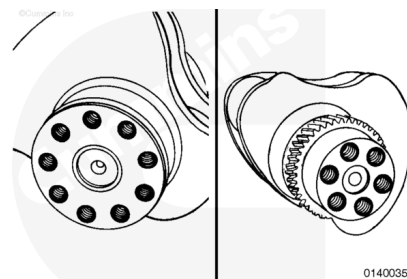
Steam clean or use hot, soapy water to clean the crankshaft and gear(s).

Use a non-metallic bristle brush to clean the oil drillings. Dry with compressed air. Make sure to blow out the threaded holes on each end of the crankshaft and the oil drillings.

⚠ CAUTION ⚠

Do not chase threads on the crankshaft. Severe engine damage can occur.

Check the threads for damage at both ends of the crankshaft.
If damage is found, replace the crankshaft.



Use a fine crocus cloth to polish the machined surfaces.

Inspect the front crankshaft gear. Refer to Procedure 001-019 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-019.html>)

Inspect the front crankshaft seal contact area. Refer to Procedure 001-023 in Section 1.

(</qs3/pubsys2/xml/en/procedures/40/40-001-023-tr.html>)

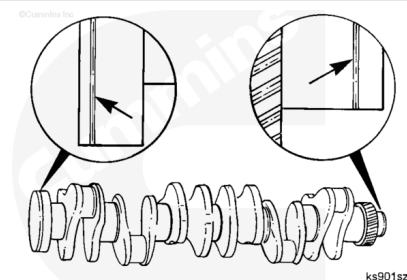
Inspect the rear crankshaft seal contact area. Refer to Procedure 001-024 in Section 1.

(</qs3/pubsys2/xml/en/procedures/40/40-001-024-tr.html>)

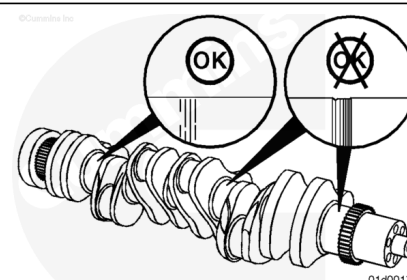
Inspect the connecting rod bearings. Refer to Procedure 001-005 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-005.html>)

Inspect the main bearings. Refer to Procedure 001-006 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-006.html>)

Note : For rear gear train engines, inspect the rear crankshaft gear. Inspect the crankshaft gear for pitting, fretting, or missing teeth. The rear crankshaft gear is not replaceable. If fretting can not be removed using abrasive pad, Part Number 3823258 or equivalent, the crankshaft must be replaced.



Check the main bearing journals and the rod bearing journals for damage or excessive wear. Minor scratches are acceptable.



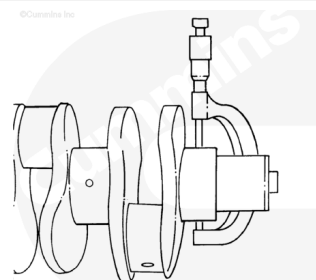
Measure

Measure the crankshaft main and connecting rod journals.

Out-of-Round: 0.050 mm [0.002 in].

Taper: 0.013 mm [0.0005 in].

Note : If crankshaft journals are not within the given specification, the crankshaft must be reground. Always grind all of the journals when one is not within specifications.

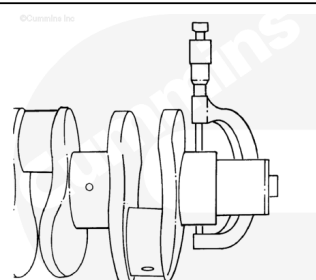


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Measure and record the diameter of the connecting rod journal on the crankshaft.

B3.9, B4.5 RGT, and B5.9 Standard Crankshaft
Connecting Rod Journal Diameter

	mm		in
Standard	68.962	MIN	2.7150
	69.013	MAX	2.7170



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B3.9, B4.5 RGT, and B5.9 Undersize
Crankshaft Connecting Rod Journal
Diameter

	mm		in	
Undersi ze	0.25 mm [0.010 in]	68.712	MIN	2.7052
	68.763	MAX	2.7072	
0.50 mm [0.020 in]	68.462	MIN	2.6954	
	68.513	MAX	2.6974	
0.75 mm [0.030 in]	68.212	MIN	2.6855	

	68.263	MAX	2.6875
1.00 mm [0.040 in]	67.962	MIN	2.6767
	68.013	MAX	2.6787

**B4.5 Standard Crankshaft Connecting Rod Journal
Diameter**

	mm		in
Standard	65.987	MIN	2.5979
	66.013	MAX	2.5989

**B4.5 Undersize Crankshaft Connecting Rod
Journal Diameter**

	mm		in	
Undersi ze	0.25 mm [0.010 in]	65.737	MIN	2.5879
	65.763	MAX	2.5889	
0.50 mm [0.020 in]	65.487	MIN	2.5779	
	65.513	MAX	2.5789	
0.75 mm [0.030 in]	65.237	MIN	2.5679	
	65.263	MAX	2.5689	
1.00 mm [0.040 in]	64.987	MIN	2.5579	
	65.013	MAX	2.5589	

Note : If the crankshaft connecting rod journals are not within the given specification, the crankshaft must be reground. Select the target undersize journal diameter and its corresponding oversize bearing. Machine the crankshaft to the maximum journal diameter specified

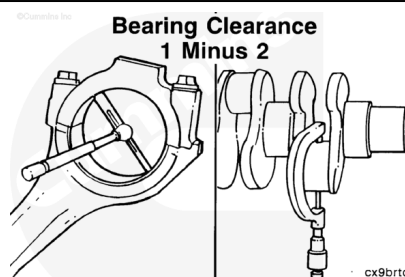
in the table above. Then finish the crankshaft to the target diameter. Always grind all of the journals when one is not within specifications.

Measure and record the diameter of the connecting rod crankshaft bore with the bearings installed. Refer to Procedure 001-014 in Section 1 for measuring the connecting rod crankshaft bore (with bearing).

(/qs3/pubsys2/xml/en/procedures/40/40-001-014.html)

Calculate the connecting rod bearing to crankshaft journal clearance.

Bearing clearance equals the connecting rod crankshaft bore (with bearing) minus the crankshaft connecting rod journal diameter.



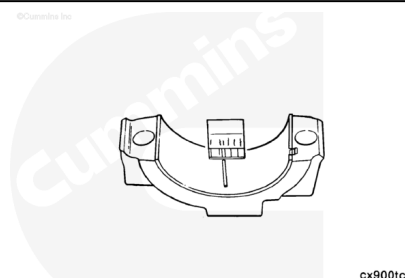
Connecting Rod to Crankshaft Bearing Clearance

mm		in
0.04	MIN	0.002
0.12	MAX	0.005

Bearing clearance can also be determined with a plastigauge during crankshaft installation.

Note : If the clearance is found to be out of specification:

1. Replace or try another set of connecting rod bearings
2. Measure and inspect the connecting rod crankshaft bore with the bearings removed. Refer to Procedure 001-014 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-014.html)



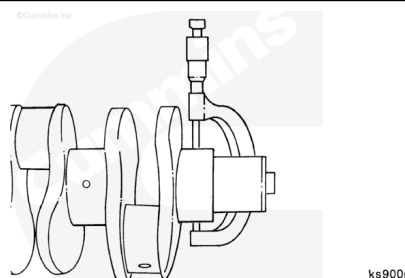
Measure and record the diameter of the main bearing journal on the crankshaft.

Standard Crankshaft Main Bearing Journal Diameter

	mm		in
Standard	82.962	MIN	3.2662
	83.013	MAX	3.2682

Undersize Crankshaft Main Bearing Journal Diameter

	mm		in
--	----	--	----



Undersize	0.25 mm [0.010 in]	82.712	MIN	3.2564
	82.763	MAX	3.2584	
0.50 mm [0.020 in]	82.462	MIN	3.2465	
	82.513	MAX	3.2485	
0.75 mm [0.030 in]	82.212	MIN	3.2367	
	82.263	MAX	3.2387	
1.0 mm [0.040 in]	81.962	MIN	3.2268	
	82.013	MAX	3.2289	

Note : If the crankshaft connecting rod journals are not within the given specification, the crankshaft must be reground. Select the target undersize journal diameter and its corresponding oversize bearing. Machine the crankshaft to the maximum journal diameter specified in the table above. Then finish the crankshaft to the target diameter. **Always** grind all of the journals when one is not within specifications.

Install the main bearing caps with the upper and lower main bearings. Refer to Procedure 001-006 (Bearings, Main) in Section 1 for main bearing cap installation.

(/qs3/pubsys2/xml/en/procedures/40/40-001-006.html)

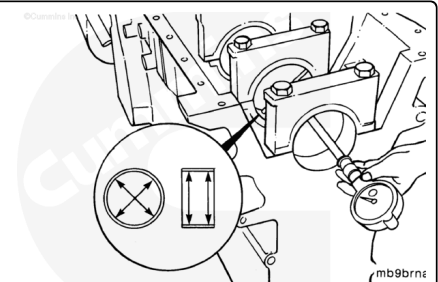
Tighten the main bearing cap capscrews.

Torque Value: 176 n•m [130 ft-lb]

Measure the main bearing bore with the bearings installed.

Calculate the main bearing to crankshaft journal clearance.

Bearing clearance equals the main bearing crankshaft bore (with bearing) minus the crankshaft main bearing journal diameter.



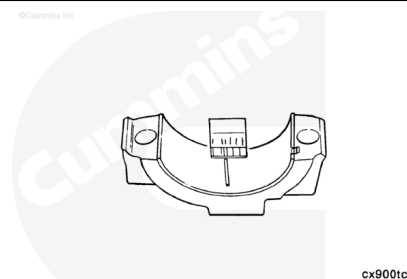
Main Bearing Bore to Crankshaft Bearing Clearance

mm		in
0.04	MIN	0.002
0.12	MAX	0.005

Bearing clearance can also be determined with a plastigauge during engine assembly.

Note : If the clearance is found to be out of specification:

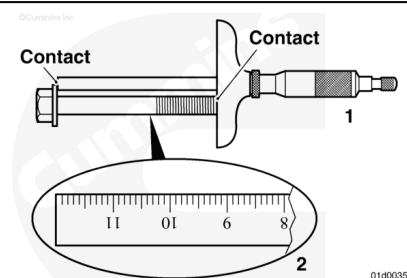
1. Replace or try another set of main bearings
2. Measure and inspect the main bearing bore with the bearings removed. Refer to Procedure 001-026 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-026.html>)



Main Bearing Capscrew Reuse Measurement

⚠ CAUTION ⚠

This step must be completed on B4.5 RGT engines. Failure to check the main bearing capscrew against reuse guidelines can result in severe engine damage.



To check if a main bearing capscrew can be reused, the length **must** be measured by performing the following:

For each main bearing capscrew that has been removed, measure the length from underneath the head of the capscrew to the tip of the capscrew, as illustrated, using one of two methods.

1. A depth micrometer (preferred method for accuracy)
2. A machinist's rule.

If the measurement is above the maximum specification, the main bearing capscrew **must** be replaced.

Main Bearing Underhead Capscrew Length

mm		in
120.00	MAX	4.724

Check the thrust bearing surfaces for damage or excessive wear. Minor scratches are acceptable. Use a fine crocus cloth to polish the machined surfaces.

The thrust bearing surfaces are located at:

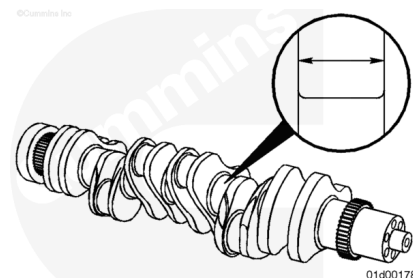
For 4 cylinder engines - The number 4 crankshaft main bearing journal.

For 6 cylinder engines - The number 6 crankshaft main bearing journal.

Measure the thrust distance.

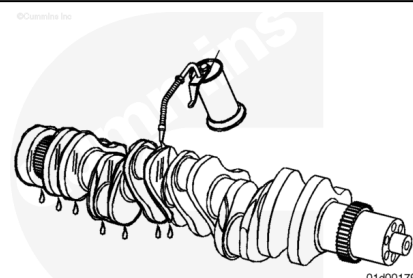
Thrust Distance		
mm		in
37.475	MIN	1.4754
37.576	MAX	1.4794

Note : Oversize thrust bearings are available if the thrust distance is **not** within specifications. Oversize thrust bearings of 0.25 and 0.50 mm [0.010 to 0.020 in] are available.



Use a light preservative oil to lubricate the crankshaft to prevent rust.

Note : If the crankshaft is **not** going to be used immediately, use a heavy preservative oil. Protect the crankshaft with a cover to prevent dirt from sticking to the oil.

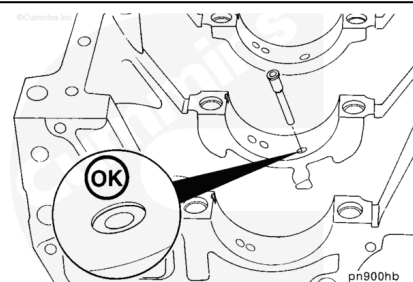


Initial Check

For engines equipped with drillings for saddle jet piston cooling nozzles/plugs, verify that:

- engines originally equipped with saddle jet piston cooling nozzles, the piston cooling nozzles are installed and free from debris/damage
- engines equipped with J-jet piston cooling nozzles, the saddle jet piston cooling nozzle drillings are plugged.

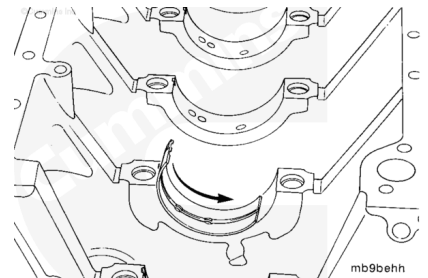
Note : Some engine do are **not** contain drillings for saddle jet piston cooling nozzles/plugs.



Install

Make sure the backsides of the bearings are clean and free of debris before installing the upper main bearings into the block.

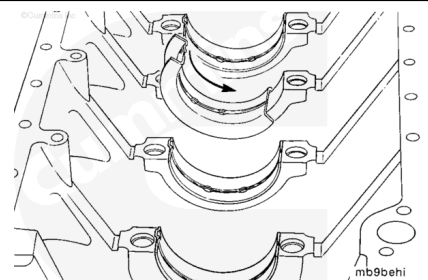
Make sure to align the tangs of the bearings with tangs on the main bearing block saddles.



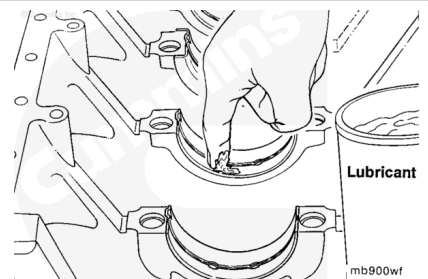
Install the upper crankshaft thrust bearing.

4 cylinder engines - The number 4 main bearing position.

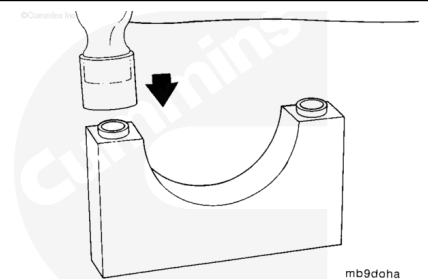
6 cylinder engines - The number 6 main bearing position.



Apply a coat of assembly lube, Part Number 3163087, to the crankshaft side of the main bearings and thrust bearing surfaces.



Check the main bearing caps to make sure the ring dowels are installed.

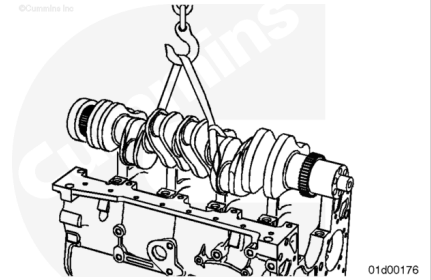


⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ CAUTION ⚠

Carefully install the crankshaft to avoid damage to the crankshaft main bearings, especially the thrust/main bearing journals. Engine life will be shortened if damage to the crankshaft occurs.

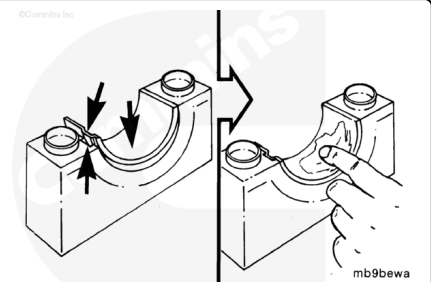


Install the crankshaft.

Note : When setting the crankshaft into the block, make sure the front crankshaft gear meshes with the lubricating oil pump gear (if not previously removed).

⚠ CAUTION ⚠

Do not lubricate the back side of the bearing that contacts the main bearing cap.



Make sure the backsides of the bearings are clean and free of debris before installing the lower main bearings into the main bearing caps.

Make sure to align the tangs of the bearings with tangs on the main bearing caps.

Note : Some engines use a thrust bearing for the upper and lower main bearing (360 degree), while other engines only have a thrust bearing for the upper main bearing (180 degree). Always replace like for like.

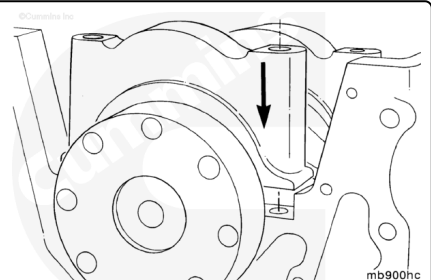
If equipped, install the lower crankshaft thrust bearing.

4 cylinder engines - The number 4 main bearing position.

6 cylinder engines - The number 6 main bearing position.

Apply a coat of assembly lube, Part Number 3163087, to the crankshaft side of the main bearings and thrust bearing surfaces.

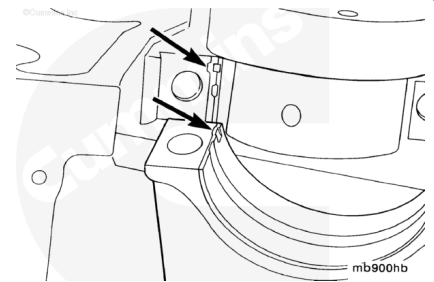
Make sure the main bear cap surfaces between the main bearing cap and block are clean and free of debris.



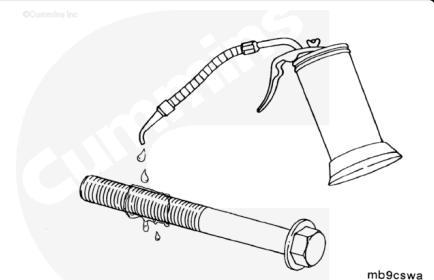
The main bearing caps are/were numbered during the removal process for their location. Number 1 starts with the front of the block.

Note : The caps must be installed so the numbers on the caps match the bearing saddle in the block. The lock tangs in the main bearing saddle and bearing cap must be on the same side.

Install the main bearing caps. Make sure to align the ring dowels on the main bearing cap with the corresponding drillings in the cylinder block.



Lubricate the main bearing capscrew threads and underside of the head with clean engine oil.

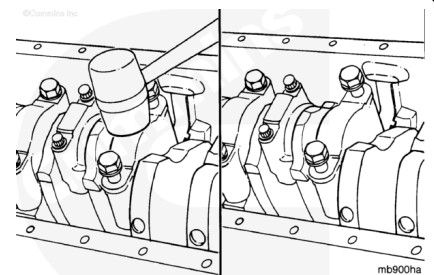


Gently tap the main bearing cap into position with a plastic or rubber mallet.

When seated, install the main bearing capscrews and tighten.

Torque Value: 50 N·m [37 ft-lb]

Do **not** tighten to the final torque value at this time. Final torque should be applied after all main bearing caps are installed.



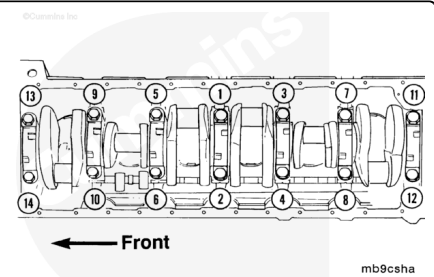
Note : The sequence to the right is for a 6 cylinder engine. For a 4 cylinder engine, use the same sequence for the 5 main bearing caps.

Tighten the capscrews evenly and in sequence. Perform each step to all capscrews before performing the next step.

B3.9, B4.5, and B5.9 Engines:

Torque Value:

1. 60 N·m [44 ft-lb]
2. 90 N·m [66 ft-lb]
3. Turn all capscrews through 90 degrees.



⚠ CAUTION ⚠

For B4.5 RGT engines, there is a different torque procedure for new and previously installed main bearing capscrews. Failure to use the correct torque value can result in engine damage.

B4.5 RGT Engines:

Torque Value:

Previously Installed Main Bearing Capscrews

1. 60 n•m [44 ft-lb]
2. 80 n•m [59 ft-lb]
3. Rotate 90 degrees.

Torque Value:

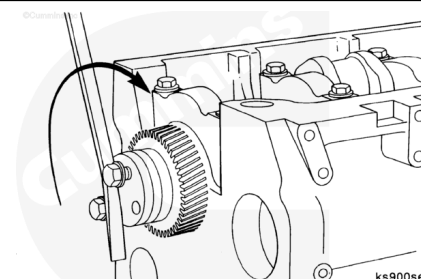
New Main Bearing Capscrews

1. 120 n•m [89 ft-lb]
2. Loosen completely.
3. 60 n•m [44 ft-lb]
4. 85 n•m [63 ft-lb]
5. Rotate 120 degrees.

The crankshaft **must** rotate freely after installing the main bearing caps.

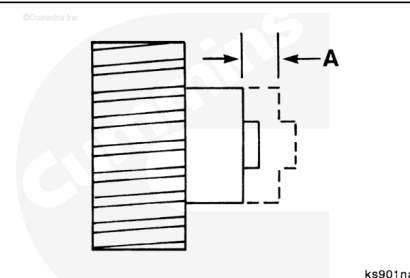
If the crankshaft does **not** rotate freely:

1. Check if the crankshaft is contacting one of the connecting rods
2. Check if the correct main bearing caps were installed correctly
3. Check if the main bearing cap ring dowels or mounting surfaces were damaged during installation
4. Check if the correct main bearings were installed.



Note : The dimensions of the thrust bearing and crankshaft journal determine end play.

Measure the crankshaft end play with a dial indicator assembly, Part Number 3824564 and magnetic base, Part Number 3377399.



Crankshaft End Play		
mm		in
0.102	MIN	0.004
0.432	MAX	0.017

If the crankshaft end play is not within specification:

1. If the crankshaft end play is below specification, check if there are any obstructions limiting the crankshaft's travel (lubricating oil pump, connecting rod, etc.)
2. If the crankshaft end play is above specification, inspect the crankshaft thrust bearing surface. Also check if the correct thrust bearing(s) were installed.

Note : Oversize thrust bearings are available if the end play is not within specifications. Oversize thrust bearings of 0.25 to 0.51 mm [0.010 to 0.020 in] are available.

Finishing Steps

- Install the connecting rod caps. Refer to Procedure 001-005 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-005.html)
- If equipped, install the J-jet piston cooling nozzles. Refer to Procedure 001-046 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-046-tr.html)
- For front gear train engines, install the front gear housing. Refer to Procedure 001-033 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-033-tr.html)
- For rear gear train engines, install the rear gear housing. Refer to Procedure 001-034 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-034-tr.html)



Service Tip: With the engine still on the engine stand and upside down, the camshaft should be installed. This will keep the tappets suspended so that the camshaft can be installed. This eliminates the need to use tappet replacement kit, Part Number 3822513, to raise the tappets. Once the camshaft is installed, rotate the engine upright so the tappets will contact the camshaft.

- Install the fuel transfer pump. Refer to Procedure 005-045 in Section 5. (/qs3/pubsys2/xml/en/procedures/40/40-005-045-tr.html)
- Install the camshaft. Refer to Procedure 001-008 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-

008-tr.html)

- For rear gear train engines, install the camshaft gear. Refer to Procedure 001-012 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-012-tr.html)
- Install the fuel injection pump. Use the following procedure for rotary fuel injection pumps. Refer to Procedure 005-014 in Section 5. For inline fuel injection pumps use the following procedure.
(/qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html) Refer to Procedure 005-012 in Section 5.
(/qs3/pubsys2/xml/en/procedures/40/40-005-012-tr.html)
- If equipped, install the air compressor. Refer to Procedure 012-014 in Section 12.
(/qs3/pubsys2/xml/en/procedures/40/40-012-014-tr.html)
- If equipped, install or connect any driven accessories (i.e., hydraulic pump)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html)
- If equipped, install the balancer. Refer to Procedure 001-004 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-004.html)
- If equipped install the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-089-tr.html)
- Install the lubricating oil pan and gasket. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html)
- For front gear train engines, install the rear seal and rear seal carrier. Refer to Procedure 001-067
(/qs3/pubsys2/xml/en/procedures/40/40-001-067-tr.html) and Refer to Procedure 001-104 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-104-tr.html)
- Install the flywheel housing. Refer to Procedure 016-006 in Section 16. (/qs3/pubsys2/xml/en/procedures/40/40-016-006-tr.html)
- For rear gear train engines, install the rear seal. Refer to Procedure 001-024.

(/qs3/pubsys2/xml/en/procedures/40/40-001-024-tr.html)

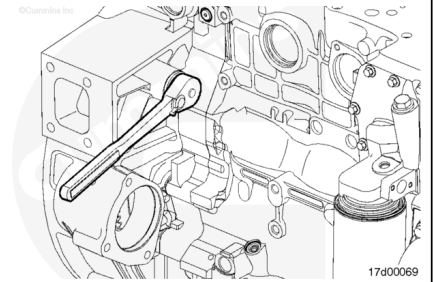
- Install the flywheel or flexplate. Refer to Procedure 016-005 (/qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html) or Refer to Procedure 016-004 in Section 16. (/qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html)
- Install the front cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-031-tr.html)
- Install the vibration damper/crankshaft pulley. Use the following procedure if equipped with a viscous damper. Refer to Procedure 001-052 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html) If equipped with a rubber damper use the following procedure. Refer to Procedure 001-051 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html) Use the following procedure if equipped with a crankshaft pulley **only**. Refer to Procedure 001-022 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-022-tr.html)
- If required, install the fan hub pulley. Refer to Procedure 008-039 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html)
- Install the cooling fan drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html)
- Install the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/40/40-004-014-tr.html)
- Install the rocker levers. Refer to Procedure 003-008 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html)

Rotation Check

With the engine fully assembled, check that the engine rotates freely using barring tool, Part Number 3824591.

Insert the barring tool into the flywheel housing and engage the flywheel/flexplate ring gear. The crankshaft can then be rotated by hand, use a 1/2- inch drive ratchet or breaker bar.

If the engine does **not** rotate freely, check for any external obstructions. If no obstructions are found, remove the oil pan and look for internal damage.



- Install the engine. Refer to Procedure 000-002 in Section 0. (</qs3/pubsys2/xml/en/procedures/40/40-000-002.html>)
- Fill the lubricating oil pan. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>)
- Operate the engine. Check for leaks and proper engine operation.

